Occlusion-Robust Human Gait Recognition Using Deep Learning: A Master's Project Framework

1. Recommended Narrow Topic

1.1 Core Research Focus

1.1.1 Primary Title: "Deep Learning-Based Gait Recognition Robust to Partial Occlusion in Surveillance Scenarios" The proposed research addresses **one of the most persistent barriers to practical gait recognition deployment**: the severe performance degradation that occurs when subjects are partially occluded by environmental obstacles, other pedestrians, or scene elements. While laboratory-validated systems routinely achieve **>95% Rank-1 accuracy** on clean benchmarks like CASIA-B, operational performance in crowded public spaces often collapses to **<60%** due to occlusion effects (ECVA) . This discrepancy between controlled evaluation and real-world utility represents a critical research gap that directly limits technology adoption in security-critical applications.

The title deliberately positions the work at the intersection of three essential dimensions: **deep learning methodology** (addressing constraint #7), **occlusion robustness as core technical challenge** (addressing constraint #4 for novel results), and **surveillance as primary application domain** (addressing constraint #9 for relevance). The explicit mention of "partial occlusion" signals realistic problem framing rather than idealized conditions, distinguishing this work from foundational gait recognition studies that assume complete visibility. The surveillance context is particularly apt because it presents **occlusion as a pervasive operational condition rather than an exceptional edge case** — a fundamental reframing that motivates architectural innovations integrating occlusion awareness throughout the recognition pipeline.

1.1.2 Problem Statement: Addressing Recognition Degradation When Subjects Are Partially Hidden The core technical problem decomposes into three interconnected challenges that must be solved in concert:

Challenge	Description	Consequence of Inadequate Solution
Occlusion Detection	Accurately identifying which body regions are visible vs. occluded in each frame	Misclassification of occluded regions as valid features corrupts recognition decisions
Feature Extraction	Learning representations that emphasize reliable visible regions while suppressing unreliable occluded regions	Network wastes capacity processing noise; discriminative information from visible regions underutilized
Temporal Fusion	Aggregating information across frames where occlusion state varies dynamically	Static fusion strategies fail to exploit temporal continuity; occluded frames contaminate sequence representations

Quantitative evidence from recent benchmarks substantiates the severity of this problem. On

OccCASIA-B, state-of-the-art methods like **GaitPart experience 15-30 percentage point accuracy drops** under moderate occlusion, with severe occlusion causing **>50% degradation** (ECVA) . The **GaitMoE** study explicitly documents that "set-based and temporal-based gait recognition methods suffer from severe degradation under the occlusion scenario, comparing to their performances in their paper on holistic CASIA-B" (ECVA) . These figures translate directly into operational impacts: surveillance systems that are effectively non-functional during peak crowd periods when they are most needed.

The problem is **compounded by occlusion asymmetry** — different body regions carry differentially discriminative information, and occlusion does not affect all regions uniformly. The legs and feet, which encode characteristic gait dynamics, are frequently occluded in crowded scenes, while the upper body may remain visible but contain less distinctive motion patterns. This structural aspect of the problem motivates **part-aware processing** that can adaptively weight contributions based on both visibility and discriminative importance.

1.1.3 Novelty Angle: Integration of Attention Mechanisms with Spatial-Temporal Feature Fusion for Occlusion-Aware Gait Identification

The proposed novelty centers on **systematic integration of three architectural innovations**:

1. **Learnable occlusion detection modules** that automatically identify and localize occluded regions without requiring explicit pixel-level supervision, trained through end-to-end backpropagation from recognition objectives
2. **Hierarchical attention mechanisms** operating at multiple granularities — frame-level spatial attention identifying visible regions, part-level attention weighting body segments by discriminative importance, and temporal attention selecting informative phases of the gait cycle
3. **Occlusion-conditioned spatial-temporal fusion** that propagates information across frames based on estimated reliability, exploiting gait periodicity to compensate for missing information rather than discarding occluded frames

This integration **distinguishes the approach from prior work** in several respects. The **GaitMoE** architecture addresses occlusion through **mixture-of-experts** with specialized temporal and action experts (ECVA) , while the proposed attention-fusion approach emphasizes **interpretable, modular occlusion handling** that can be diagnosed and extended. Reconstruction-based methods like **BGaitR-Net** employ **CVAE-Bi-LSTM for frame inpainting** (ScienceDirect) — computationally expensive and risking hallucination artifacts — whereas the proposed method operates directly on available features without explicit generation. The novelty extends to **occlusion-aware consistency regularization** that enforces embedding similarity across different occlusion patterns of the same subject, explicitly training for invariance rather than hoping it emerges from data augmentation.

1.2 Alignment with Project Constraints

1.2.1 Feasibility: Leverages Established Occlusion-Annotated Datasets with Pre-Processed Silhouettes

The project's feasibility rests on **three critical enabling factors**:

Factor	Specific Resource	Contribution to Feasibility
Purpose-built dataset	**OccGait** (2024): 101 subjects, 80,000+ sequences, 4 occlusion categories, 8 views (ECVA)	Eliminates 3-4 months of data collection; provides explicit occlusion annotations for supervised learning

Table 2 – continued

Factor	Specific Resource	Contribution to Feasibility
Pre-processed silhouettes	Binary silhouette sequences at 64×44 resolution, spatially aligned	Reduces preprocessing to dataset-specific adaptation rather than pipeline development from scratch
Validated training configurations	Batch size [8,16], 40K iterations, SGD with step learning rate (ECVA)	Provides proven starting point for hyperparameter search, reducing exploration space

The **OccGait dataset** is specifically designed for this research problem, with synthetic occlusion generation that preserves natural walking dynamics while enabling **precise control over occlusion type, position, and severity**. The four occlusion categories — **Non-Occlusion (NO), Carrying Occlusion (CA), Crowd Occlusion (CO), and Static Occlusion (ST)** — support fine-grained analysis of method behavior that would be impossible with undifferentiated occlusion (ECVA) . The dataset's derivation from **CASIA-B** ensures that underlying gait characteristics are representative of standard benchmarks, facilitating meaningful comparison with extensive prior literature.

The **modular architecture design** further supports feasibility by enabling **independent development and validation of components**:

```
Module 1: Data pipeline and preprocessing (Week 1)
    ↓
Module 2: Baseline recognition network without occlusion handling (Week 2)
    ↓
Module 3: Occlusion detection and attention mechanisms (Weeks 3-4)
    ↓
Module 4: Temporal fusion and system integration (Week 5)
    ↓
Module 5: Comprehensive evaluation and documentation (Week 6)
```

This decomposition ensures that **core functionality is validated early**, with clear fallback options if specific components prove challenging. The use of **PyTorch 2.0+** with `torch.compile` optimization and established attention implementations (CBAM, SE blocks) accelerates development by eliminating low-level implementation (ECVA) .

1.2.2 Timeline Suitability: 6-Week Development Cycle Achievable Through Modular Architecture Design The **1.5-month timeline** (approximately 6 weeks of intensive development) is achievable through **aggressive scope management and strategic reuse**:

Week	Primary Activities	Deliverables	Risk Mitigation
1	Environment setup; dataset download; baseline reproduction	Running data loader; training loop for standard GaitSet/GaitPart	Start dataset download immediately; verify GPU access
2	Core architecture: spatial backbone + attention mechanisms	Forward pass with attention visualization; initial training runs	Use pre-trained backbones; validate with small data subset

Table 3 – continued

Week	Primary Activities	Deliverables	Risk Mitigation
3-4	Temporal fusion; occlusion-aware training; hyperparameter optimization	Full model training to convergence; validation performance tracking	Checkpoint frequently; early stopping to prevent overfitting
5	Comprehensive evaluation; ablation studies; failure analysis	Complete result tables; attention map visualizations; timing benchmarks	Automated evaluation scripts; predefined analysis protocols
6	Paper drafting; code documentation; reproducibility packaging	Submission-ready manuscript; public repository with README	Prioritize core results; defer supplementary experiments

Critical path items are identified with specific mitigations: dataset access delays (immediate registration, institutional mirrors), GPU queue constraints (cloud backup: Google Colab Pro, Kaggle P100), and training instability (comprehensive logging, gradient clipping, learning rate warmup). The **predefined success criteria — >10% Rank-1 improvement over GaitPart baseline on OccGait CO category** — provide clear evaluation threshold without requiring exhaustive exploration.

1.2.3 Contribution Potential: Addresses Critical Gap Between Laboratory Conditions and Real-World Deployment

The research contribution is **substantial and timely** for several reasons:

- **Problem significance**: Occlusion robustness is **explicitly identified** in recent surveys as critical for "increasing the acceptance and adoption of gait recognition technology" (MDPI) , with the 2024 GaitMoE paper noting that "extensive occlusions in real-world scenarios pose challenges to gait recognition" motivated their benchmark creation (ECVA)
- **Methodological novelty**: The attention-fusion integration represents **distinct alternative** to existing mixture-of-experts and reconstruction-based approaches, with potential for **complementary strengths** in interpretability and efficiency
- **Evaluation rigor**: Systematic analysis across **occlusion types, severity levels, and cross-dataset generalization** provides template for future research and informs operational requirements
- **Open science commitment**: Public code release with reproducible benchmarks enables **community adoption and extension**, amplifying impact beyond single publication

The **practical impact** extends to immediate deployment considerations: surveillance system developers can incorporate occlusion-aware components to **maintain effectiveness during peak usage periods**, reducing false negatives that compromise security and false positives that waste operator attention.

2. Application Domain Analysis

2.1 Primary Domain: Intelligent Surveillance Systems

2.1.1 Scenario: Long-Distance Person Re-Identification in Crowded Public Spaces

The **primary deployment scenario** encompasses large-scale public space monitoring where **traditional biometric modalities face fundamental limitations**:

Modality	Operational Range	Cooperation Required	Occlusion Sensitivity	Suitability for Surveillance
Face recognition	<10 meters	Yes (pose, expression)	High (masks, angles)	Limited
Fingerprint/iris	Contact proximity	Yes (active presentation)	N/A (controlled capture)	Infeasible
Gait recognition	50-200 meters	No	Moderate (addressed by this project)	**Optimal**

Gait offers **unique advantages for surveillance**: it operates at distance without subject awareness, functions with low-resolution imagery where face recognition fails, and maintains effectiveness despite clothing changes that frustrate appearance-based methods. However, these advantages are **negated by occlusion sensitivity** in current systems — a subject clearly visible entering a plaza may become **unidentifiable when crossing near a crowd or structure**, with identity only recoverable after clearing the occlusion, if at all.

The **re-identification task** — determining whether observations at different times and locations match — **amplifies occlusion challenges** because subjects traverse complex environments with **continuously varying visibility conditions**. Effective re-identification requires **maintaining identity estimates through varying occlusion states** rather than discarding all occluded observations, demanding the **temporal continuity exploitation** central to the proposed architecture.

2.1.2 Challenge: Maintaining Recognition Accuracy Under Dynamic Occlusion Conditions

The **specific technical challenge** involves **three dimensions of dynamic variation**:

1. **Temporal dynamics**: Occlusion duration varies from **fractions of a second** (brief crowd intersection) to **multiple gait cycles** (static environmental occlusion), with transitions between occluded and visible states that must be tracked
2. **Spatial patterns**: Occlusion affects **different body regions differentially** — legs frequently occluded by low obstacles, torso by crowd at similar height, head by overhead structures — requiring **part-specific handling**
3. **Type interaction**: Real scenarios present **mixed occlusion types** — a subject may experience static occlusion from a pillar while simultaneously experiencing crowd occlusion from passing pedestrians

Current approaches fail because they **lack explicit occlusion modeling**: frame-independent processing treats occluded and visible frames equally, holistic representations are corrupted by any missing region, and temporal aggregation without reliability weighting allows occluded frames to contaminate sequence representations. The proposed **occlusion-conditioned processing** addresses each failure mode through **dynamic, input-adaptive computation**.

2.1.3 Impact: Enhanced Security Effectiveness Without Requiring Cooperative Subject Behavior

The **societal impact** of improved occlusion-robust gait recognition extends across **security application categories**:

Application	Current Limitation	Improvement from Occlusion Robustness
Threat detection	Missed identification during crowd events	Continuous tracking through occlusion maintains situational awareness
Forensic investigation	Unusable footage with partial visibility	Extraction of identity information from degraded archival video
Access control	Failure during peak traffic periods	Reliable authentication despite environmental clutter
Border security	Gaps in traveler tracking	Seamless identification across checkpoint transitions

The **non-cooperative nature** is particularly valuable for identifying **individuals actively avoiding detection** — unlike face recognition where simple avoidance of camera direction is effective, gait requires no subject awareness and is **difficult to deliberately alter without conspicuous behavior change**. The occlusion robustness contribution **removes the primary remaining barrier** to operational deployment, enabling reliable performance in realistic conditions.

2.2 Secondary Domain Considerations

2.2.1 Healthcare Monitoring: Gait Analysis for Fall Risk Assessment in Home Environments with Furniture Occlusion Beyond security, **occlusion-robust gait analysis enables healthcare monitoring** in domestic environments where **furniture and architectural elements routinely obstruct camera views**. In-home fall risk assessment systems must operate with **fixed camera placements** that create **persistent blind spots** — subjects walking behind sofas, through doorways, or around kitchen counters produce **partial observations that current analysis systems struggle to interpret** (MDPI) .

The **healthcare domain emphasizes different performance characteristics**: **longitudinal consistency** (tracking the same individual over months), **sensitivity to subtle gait changes** (early indicators of functional decline), and **tolerance for occasional identification failures** (resolvable through human review). The occlusion robustness techniques developed for surveillance **transfer directly**, with potential adaptation for **gait parameter estimation accuracy** (stride length, step time variability) rather than pure identification.

2.2.2 Forensic Investigation: Identification from Partial Footage in Criminal Incident Reconstruction **Forensic applications** present **unique requirements** that motivate specific technical extensions. Criminal incident footage is **frequently degraded**: bystander mobile devices, distant cameras, low-quality surveillance systems, with **severe, uncontrolled occlusion** from scene elements. The identification task typically involves **small-gallery matching** (specific suspect against incident footage) rather than open-set identification, potentially enabling **more aggressive exploitation of subject-specific information**.

The **legal admissibility requirements** impose stringent validation needs: **established error rates, explicit uncertainty quantification, and comprehensive audit trails**. The proposed evaluation framework — with **systematic performance characterization across occlusion types and confidence calibration** — directly supports these requirements. The **attention visualization** capability provides **interpretable evidence of decision basis**, valuable for expert testimony and building judicial confidence in algorithmic evidence.

2.2.3 Sports Analytics: Athlete Tracking During Competitive Events with Opponent Interference **Sports analytics** represents a **commercially valuable domain** with **distinct occlusion characteristics**. Competitive environments feature **predictable, rule-constrained movement patterns** that can be exploited: athletes follow position roles, maintain team formations, and exhibit **characteristic motion styles** that simplify tracking. The **opponent interference** creating occlusion is **itself analytically relevant** — defensive positioning and physical contact encode tactical information beyond pure identification.

Specific applications include **player workload estimation** from movement patterns, **tactical analysis** of positioning and interaction, and **automated highlight generation** based on exceptional individual performances. The **high frame rates** (25-60 fps) and **controlled lighting** of professional broadcasting create favorable conditions, though **rapid, unpredictable movements** of elite athletes impose demanding **temporal resolution requirements**. The occlusion challenge is **intensified by deliberate obstruction** integral to competitive strategy, requiring methods that can **identify athletes despite active blocking attempts**.

3. Publicly Available Datasets

3.1 Primary Dataset: OccGait

3.1.1 Source: Beijing Normal University - Intelligent Vision Computing Lab (BNU-IVC) The **OccGait dataset** emerges from the **Intelligent Vision Computing Laboratory at Beijing Normal University**, a research group with **established expertise in gait analysis and biometric recognition**. The dataset represents a **deliberate, systematic effort to address the occlusion gap** in available benchmarks, designed with explicit attention to occlusion types occurring in practical surveillance deployments. Institutional backing ensures **dataset quality, documentation standards, and continued maintenance**, with GitHub-based distribution supporting **community engagement and version-controlled updates** (ECVA) .

The **2024 release date** is strategically significant: the dataset has **not yet been extensively exploited** in published research, providing a **window of opportunity for establishing benchmark results and methodological baselines** before saturation. This recency ensures alignment with **current community standards** for evaluation protocols and performance metrics.

3.1.2 Access: https://github.com/BNU-IVC/OccGait The **GitHub distribution mechanism** provides **multiple practical advantages**:

- **Immediate accessibility**: No formal application procedures, institutional agreements, or waiting periods
- **Version control**: Documented release history with community-contributed extensions
- **Infrastructure integration**: Natural fit with modern research workflows and automated pipelines
- **Issue tracking**: Responsive maintenance and troubleshooting support

Researchers should **verify repository activity** (commit history, issue responses) for long-term dependency planning, though for the **1.5-month project timeline**, immediate download with local backup mitigates availability risk.

3.1.3 Characteristics

3.1.3.1 Synthetically Occluded Gait Sequences with Four Occlusion Categories The **defining characteristic** of OccGait is **systematic occlusion categorization enabling detailed performance**

analysis:

Category	Code	Simulation Method	Typical Real-World Scenario	Recognition Challenge
Non-Occlusion	NO	Baseline without artificial occlusion	Controlled environment, clear sight lines	Establish upper bound performance
Carrying Occlusion	CA	Umbrella or luggage overlay	Subject carrying objects that alter body shape	Distinguishing carrying-induced from identity-related shape variation
Crowd Occlusion	CO	Other pedestrian silhouette overlay	Busy public spaces, transportation hubs	Dynamic, unpredictable 遮挡; potential confusion between target and occluder motion
Static Occlusion	SO	Environmental object (plant, chair) overlay	Pillar obstruction, furniture blocking	Persistent elimination of specific body regions; requires compensation from visible parts

The **synthetic generation methodology** — applying controlled 遮挡 masks to base CASIA-B silhouettes — **preserves ground-truth identity labels** while enabling **precise occlusion parameter control**. This supports **systematic variation studies** (occlusion position, size, duration) that inform operational requirements and system design decisions (ECVA) .

3.1.3.2 Dataset Scale and Structure

Attribute	Specification	Implication for Research
Subjects	101 identities	Sufficient for deep learning training without overfitting; manageable computational requirements
Sequences	80,000+ total	Extensive data for robust statistical evaluation; supports data augmentation strategies
Views	8 camera viewpoints	Cross-view evaluation; analysis of view-occlusion interaction effects
Resolution	1920×1080 video → 64×44 silhouettes	High-quality source enables resolution robustness studies; standardized silhouette format
Capture	Indoor laboratory, controlled lighting	Isolates occlusion effects from confounding covariates

The **multi-view structure** (0°, 45°, 315° physical cameras with synthesized views through walking direction variation) enables **investigation of view-invariant recognition under occlusion** — particularly relevant for surveillance where camera placement cannot be controlled (ECVA) .

3.1.3.3 Standard Evaluation Protocol The **recommended evaluation configuration** specifies:

- **Training set**: First 74 subjects (IDs 001-074)
- **Testing set**: Remaining 50 subjects (IDs 075-124)
- **Gallery**: NM01 and NM02 sequences (non-occluded, first two captures)

- **Probes**: NM03-NM04, CA01-CA02, CO01-CO02, ST01-ST02 (all conditions)

This **subject-independent, condition-stratified split** ensures that reported results reflect **generalization to unseen individuals and novel occlusion patterns**, not memorization of training data. The **clear gallery/probe definition** supports **reproducible comparison** across studies (ECVA) .

3.2 Alternative Dataset: OccCASIA-B

3.2.1 Base Dataset: CASIA Gait Database Dataset B **OccCASIA-B** extends the **most widely used gait recognition benchmark**, providing **continuity with extensive prior literature** and **cross-dataset validation opportunity**. The base **CASIA-B dataset** contains:

Attribute	Specification
Subjects	124 (93 male, 31 female)
Views	11 angles (0°-180°, 18° increments)
Conditions	Normal (NM), Bag-carrying (BG), Coat-wearing (CL)
Sequences	6 NM, 2 BG, 2 CL per subject-view combination
Total	~13,640 sequences

The **multi-condition structure** enables evaluation of **occlusion robustness under appearance variation** — bag-carrying and coat-wearing introduce **shape changes that interact with occlusion effects**, creating more challenging recognition scenarios (ECVA) .

3.2.2 Source and Access

Aspect	Details
Source	Institute of Automation, Chinese Academy of Sciences (CASIA)
Distribution	Center for Biometrics and Security Research (CBSR)
URL	http://www.cbsr.ia.ac.cn/iCPR/gait/Gait%20Databases.html
Access	Academic registration with release agreement
Processing time	Typically 3-5 business days

The **formal access procedure** — requiring signed agreement prohibiting redistribution and commercial use without authorization — is **standard for research datasets** and does not impede legitimate academic research. **Immediate initiation recommended** to avoid timeline delays.

3.2.3 Specifications

3.2.3.1 Occlusion Simulation Parameters The **OccCASIA-B extension** applies **controlled synthetic occlusion** to CASIA-B silhouettes:

Parameter	Setting	Purpose
Occlusion probability	0.6	Ensures substantial occlusion exposure without overwhelming training
Scenario types	4 (NO, CO, SO, DO)	Correspondence with OccGait categories for cross-dataset consistency
Position variation	Up, down, left, right (for DO)	Systematic coverage of spatial occlusion patterns
Split compatibility	Standard 74/50 train/test	Direct comparison with non-occluded CASIA-B results

The **synthetic generation enables precise parameter control** — occlusion size, position, and timing can be systematically varied to **characterize performance degradation curves** and **identify operational thresholds** (ECVA) .

3.2.3.2 Pre-extracted Silhouette Format CASIA-B provides **binary silhouette sequences** through **background subtraction and morphological cleaning**, with:

- **Size normalization**: Consistent dimensions across subjects and conditions
- **Temporal alignment**: Frame sequences synchronized to gait cycle
- **Quality validation**: Manual inspection and correction for major artifacts

This **standardized format eliminates preprocessing variability** that could confound method comparison, though researchers should **verify silhouette quality** for their specific experimental conditions.

3.3 Supplementary Dataset: NTU RGB+D (Adapted)

3.3.1 Source: Nanyang Technological University The **NTU RGB+D dataset** — while designed for **general action recognition** — provides **valuable complementary resources** for gait-specific research:

Attribute	Specification	Gait Research Utility
Scale	56,880 action sequences, 60 action classes, 40 subjects	Large-scale training data; cross-dataset generalization evaluation
Modalities	RGB video, depth maps, 3D skeletons	Skeleton-based feature development; multi-modal fusion investigation
Annotation	3D joint positions (25 joints)	Ground-truth pose for occlusion recovery validation; structural prior exploitation

The **"walking" action subset** — extracted from the broader action set — provides **sufficient gait-specific data for meaningful experimentation** (ECVA) .

3.3.2 Application: Skeleton-Based Gait Feature Extraction for Occlusion Recovery The **3D skeletal data** enables **investigation of complementary recognition approaches**:

Approach	Strength	Occlusion Robustness Mechanism
Silhouette-based	Rich shape detail; established benchmarks	Part-aware attention; temporal completion
Skeleton-based	Explicit structure; invariant to clothing	Joint position inference from visible limbs; kinematic constraints
Multi-modal fusion	Combined strengths; redundancy	Cross-modal compensation when one modality degraded

The **skeletal representation's explicit structure** provides **strong priors for occlusion recovery**: even with substantial joint occlusion, **visible limb segments constrain plausible configurations** of missing joints through **anatomical connectivity and range-of-motion limits** (ECVA) .

3.3.3 Utility: Cross-Validation of Pose Estimation Robustness NTU RGB+D serves **critical methodological function** for **validating pose estimation components** that may integrate into occlusion-handling pipelines:

- **Ground-truth skeletons** enable **quantitative error analysis** of pose estimation from occluded silhouettes
- **Multi-view capture** supports **view effect analysis** on estimation accuracy
- **Large scale** enables **training of robust pose estimators** for downstream gait analysis

This **diagnostic capability** is **essential for systematic improvement** of pose-based occlusion handling, identifying **which estimation errors most degrade recognition** and guiding **targeted architectural refinements**.

4. Image Processing Pipeline

4.1 Preprocessing Stage

4.1.1 Background Subtraction and Silhouette Extraction

4.1.1.1 Gaussian Mixture Models for Dynamic Background Modeling For **new video capture** (beyond pre-processed datasets), **Gaussian Mixture Models (GMM)** provide **statistically principled background subtraction**:

The GMM represents **each pixel's color distribution as a mixture of Gaussian components**, enabling **robust classification** of new observations as **background (matching established component)** or **foreground (novel appearance)**. For gait applications:

Parameter	Typical Setting	Rationale
Components per pixel	3-5	Capture repetitive background motion (foliage, screens)
Learning rate	0.005-0.01	Balance adaptation speed against foreground contamination

Table 13 – continued

Parameter	Typical Setting	Rationale
Shadow detection	Enabled	Prevent silhouette fragmentation from cast shadows
Post-processing	Opening (3×3) → Closing (5×5)	Remove noise; fill holes without merging objects

The **GMM's probabilistic formulation** naturally handles **gradual illumination changes** that would challenge simpler background models, supporting **deployment in less controlled environments** than laboratory conditions (ECVA) .

4.1.1.2 Morphological Operations for Noise Reduction **Morphological filtering** cleans binary foreground masks through **structuring element operations**:

Operation	Sequence	Effect
Erosion	First	Remove isolated noise pixels; shrink boundaries
Dilation	Second	Restore eroded regions; fill small interior holes
Opening	Erosion → Dilation	Remove small bright noise; smooth protrusions
Closing	Dilation → Erosion	Fill dark holes; connect nearby regions

For **gait silhouettes**, **vertical structuring elements** are sometimes preferred to **preserve elongated human form** while **suppressing horizontal noise streaks** from camera jitter. **Parameter validation** through **visual inspection and downstream recognition performance** is essential — **over-aggressive filtering eliminates thin limb structures** critical for discrimination (ECVA) .

4.1.1.3 Contour Detection and Region-of-Interest Isolation **Connected component analysis** identifies **distinct foreground regions**, with **largest valid region** selected as subject silhouette:

Selection Criterion	Threshold	Purpose
Minimum area	500-1000 pixels	Eliminate distant subjects, noise clusters
Maximum area	Image-dependent	Flag merged subjects, background errors
Aspect ratio	1.5-4.0 (height/width)	Reject non-human shapes

For **occlusion scenarios with multiple foreground regions**, **additional logic** is required: **motion trajectory consistency** (select region nearest previous position), **shape analysis** (human-like contour curvature), or **detection network integration** (upstream person detector confidence) (ECVA) .

4.1.2 Normalization and Standardization

4.1.2.1 Size Normalization to Fixed Dimensions **Standard dimensions** for gait recognition:

Resolution	Use Case	Trade-off
64×44	GaitMoE default (ECVA) ; efficient processing	Moderate detail; optimized for horizontal pooling
64×64	Symmetric; common in prior work	Slightly higher detail; less part-optimized
128×128	Higher fidelity; larger model capacity	Increased computation; diminishing returns for silhouettes

Normalization **preserves aspect ratio through padding** rather than distortion, maintaining **body proportion information** that may be discriminative. **Centroid alignment** to fixed image position **reduces translation variability** (ECVA) .

4.1.2.2 Temporal Alignment Using Gait Cycle Detection **Gait cycle detection** enables **phase-normalized analysis**:

Method	Implementation	Robustness Consideration
Autocorrelation of silhouette width	Peak detection in width time series	Fails with severe occlusion disrupting width pattern
Lower-body motion analysis	Foot position tracking from silhouette bottom	Requires leg visibility; compensates with upper-body features
Frequency domain analysis	FFT of principal component projections	Handles noise through spectral averaging; slower

For **occlusion-robust recognition**, **partial cycle handling** is critical: **padding** (replicate available frames), **adaptive temporal pooling** (learned importance weighting), or **cycle completion** (generative infilling of missing phases) (ECVA) .

4.1.2.3 Illumination Invariant Transformation While **silhouettes are inherently illumination-invariant**, **pre-silhouette processing** benefits from normalization:

Technique	Application	Effect
Histogram equalization	Grayscale before thresholding	Enhance contrast for reliable threshold selection
Color space conversion (HSV, YCbCr)	Separate luminance from chrominance	Process more stable channels for background modeling
Photometric normalization	Compensate for illumination direction	Reduce shadow effects on silhouette boundary

These transformations **improve silhouette quality consistency** across varying capture conditions (ECVA) .

4.2 Occlusion Handling Techniques

4.2.1 Occlusion Region Detection

4.2.1.1 Frame Differencing for Motion Segmentation **Frame differencing** identifies **regions of change** with **anomalous patterns indicating occlusion**:

Pattern	Interpretation	Occlusion Type Indicator
Expected body boundary motion	Normal walking	No occlusion
Extended difference beyond body	Additional moving object	Crowd occlusion
Suppressed difference in body region	Static object blocking motion	Static occlusion
Fragmented, inconsistent differences	Detection failure	Detection occlusion

Temporal integration (accumulation over multiple frames) **improves robustness** to **brief stillness** and **temporary ambiguity** (ECVA) .

4.2.1.2 Edge Continuity Analysis for Occlusion Boundary Identification **Silhouette boundary analysis** exploits **predictable human contour properties**:

Feature	Natural Body Characteristic	Occlusion Indicator
Curvature continuity	Smooth, anatomically constrained curves	Abrupt changes, straight segments
Topology	Single closed contour	Multiple contours, gaps
Termination pattern	Gradual limb narrowing	T-junctions with occluding object

Edge continuity metrics — how well boundary can be traced — localize occlusion boundaries for targeted processing (ECVA) .

4.2.1.3 Optical Flow-Based Motion Inconsistency Detection **Dense motion estimation** reveals **occlusion through flow field characteristics**:

Flow Pattern	Interpretation	Computational Cost
High uncertainty regions	No visible texture for matching	Moderate (sparse methods)
Vector inconsistency with neighbors	Motion discontinuity at occlusion boundary	Higher (dense estimation)

Table 21 – continued

Flow Pattern	Interpretation	Computational Cost
Magnitude discontinuities	Appearance/disappearance of occluded regions	Highest (deep flow methods)

Farneback optical flow provides **reasonable accuracy-efficiency trade-off** for silhouette inputs; **RAFT/PWC-Net** offer **improved accuracy** for **original video analysis** at **substantially higher cost** (ECVA) .

4.2.2 Inpainting and Reconstruction

4.2.2.1 Partial Convolution-Based Missing Region Completion **Partial convolution — normalization over valid inputs only — naturally accommodates variable occlusion**:

```
Standard convolution: output = W * (X ⊙ M) / (sum(M) + ε)
Partial convolution:   output = W * (X ⊙ M) / sum(M)  [if sum(M) > threshold]
                       mask_update = conv(M)  [progressive filling tracking]
```

Progressive mask update enables **deep networks to handle arbitrarily large occlusions** through **iterative refinement**. Training with **random masking strategies** simulates **diverse occlusion patterns** for **generalization** (ECVA) .

4.2.2.2 Generative Adversarial Network for Plausible Silhouette Synthesis **Conditional GAN formulation** for **controlled completion**:

Component	Function	Training Objective
Generator	Produce completion from (occluded_silhouette, occlusion_mask)	Adversarial loss + reconstruction loss + identity preservation loss
Discriminator	Distinguish real vs. completed silhouettes	Binary classification with gradient penalty for training stability

Multi-task training with **recognition objective** ensures **generated completions preserve identity information**, not merely **visual plausibility** (ScienceDirect) .

4.2.2.3 Skeleton-Guided Structural Reconstruction **Explicit structure exploitation** for **anatomically constrained completion**:

Stage	Process	Output
1. Visible joint estimation	Pose detection from available silhouette regions	Partial 2D/3D joint positions
2. Occluded joint inference	Kinematic constraints, temporal continuity, learned priors	Complete skeleton estimate
3. Silhouette generation	Shape model or learned renderer from skeleton	Completed silhouette

Strong anatomical priors — **joint connectivity, range of motion, limb proportion constraints** — **reduce completion ambiguity** compared to **unconstrained image generation** (ECVA) .

4.3 Feature Representation Generation

4.3.1 Gait Energy Image (GEI) Construction

GEI aggregates silhouettes across gait cycle:

$$\mathrm{GEI}(x, y) = \frac{1}{T} \sum_{t=1}^{T} S_t(x, y)$$

Property	Effect	Occlusion Modification
Temporal averaging	Suppress noise; emphasize consistent shape	Exclude or downweight occluded frames
Phase alignment	Normalize to canonical gait cycle	Handle partial cycles through padding/adaptive pooling
Dimensionality reduction	Single image from sequence	Loss of temporal dynamics; complement with sequence models

Occlusion-aware GEI: **weighted averaging by estimated frame reliability**, or **separate GEI construction for visible parts only** (ECVA) .

4.3.2 Gait Entropy Image (GEnI) for Dynamic Information

GEnI encodes temporal variation:

$$\mathrm{GEnI}(x, y) = -\sum_{b} p_b(x, y) \log p_b(x, y)$$

where $p_b(x, y)$ is **empirical probability of intensity bin** b **at pixel** (x, y) **across cycle**.

Region Characteristic	GEI Response	GEnI Response
Static (head, torso)	High intensity (frequently occupied)	Low entropy (stable)
Dynamic (legs, arms)	Moderate intensity (intermittently occupied)	High entropy (variable)

Combined GEI+GEnI provides **two-channel representation** encoding **both shape and dynamics** (ECVA) .

4.3.3 Frame-Level Sequence Preservation for Temporal Models

Deep sequence models — **ConvLSTM, TCN, Transformers** — **operate directly on frame sequences**:

Architecture	Temporal Modeling	Occlusion Handling Capability
ConvLSTM	Recurrent, spatial-preserving	Gating mechanisms can learn to ignore occluded frames
TCN	Dilated convolution, parallelizable	Receptive field design for cycle-scale dependencies
Transformer	Self-attention, global pairwise	Explicit occlusion masking in attention computation

Frame-level processing enables dynamic reliability weighting — learned rather than hand-crafted — for **adaptive occlusion handling** (ECVA) .

5. Deep Learning Architecture

5.1 Backbone Network Design

5.1.1 Spatial Feature Extraction

5.1.1.1 ResNet-50 or EfficientNet-B0 for Silhouette Encoding

Architecture	Parameters	Strengths	Modification for Silhouettes
ResNet-50	25.6M	Proven performance; extensive pretraining; residual connections	Single-channel input; or replicate to 3 channels
EfficientNet-B0	5.3M	Better accuracy-efficiency tradeoff; compound scaling	Same input adaptation; potentially preferable for efficiency

Pretrained ImageNet weights provide **useful initialization despite domain difference — early layers extract generic edges and textures transferable to silhouettes** (ECVA) .

5.1.1.2 Spatial Attention Modules (CBAM, SE Blocks) for Discriminative Region Focus

Module	Mechanism	Computational Overhead	Occlusion Integration
SE (Squeeze-and-Excitation)	Channel attention: global pooling → FC → sigmoid	~10% parameters	Channel-wise reweighting; indirect spatial effect
CBAM	Sequential channel + spatial attention	~20% parameters	Explicit spatial focus; can modulate by occlusion mask

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Module	Mechanism	Computational Overhead	Occlusion Integration
Custom occlusion-attention	Learned visibility estimation → multiplicative masking	Variable	Direct occlusion-aware feature suppression

CBAM's spatial attention provides **natural integration point** for **explicit occlusion handling — occlusion mask can guide or replace learned spatial attention** (ECVA) .

5.1.1.3 Part-Based Feature Learning (Head, Torso, Leg Regions) **Horizontal partitioning** for **part-specific processing**:

Part	Vertical Position	Discriminative Information	Occlusion Vulnerability
Head	Top 15%	Limited; mainly scale reference	Low (usually visible)
Torso	15-45%	Body proportions; arm swing	Moderate
Upper legs	45-70%	Gait dynamics; stride characteristics	High (frequent occlusion)
Lower legs	70-100%	Foot strike; push-off mechanics	Very high (most occluded)

Adaptive part weighting — learned or based on visibility estimation — enables recognition from available parts when others occluded (ECVA) .

5.1.2 Temporal Modeling

5.1.2.1 ConvLSTM or Bi-LSTM for Sequence Dynamics

Variant	Directionality	Memory	Occlusion Handling
ConvLSTM	Unidirectional	Convolutional state (spatial structure preserved)	Forget gate can learn to discard occluded frames
Bi-ConvLSTM	Bidirectional	Forward + backward states	Future context improves occluded frame interpretation

Bidirectional processing particularly valuable for gait — **gait cycle symmetry enables forward-backward consistency constraints** (ECVA) .

5.1.2.2 Temporal Convolutional Networks (TCN) for Variable-Length Sequences **TCN advantages for gait**:

Property	Benefit	Implementation
Parallelizable computation	Faster training than RNNs	Dilated causal convolutions
Stable gradients	No vanishing gradient problem	Residual connections, weight normalization
Adjustable receptive field	Match to gait cycle duration	Exponential dilation: 1, 2, 4, 8, $\cdots$

Causal structure for **online recognition**; **non-causal** for **offline batch processing with improved accuracy** (ECVA) .

5.1.2.3 Transformer Encoders for Long-Range Dependency Capture **Self-attention for global temporal relationships**:

Component	Function	Occlusion Adaptation
Multi-head self-attention	Compute pairwise frame affinities	Mask attention from/to occluded frames
Position encoding	Maintain temporal order information	Learned or sinusoidal; robust to missing positions
Feed-forward network	Transform attended features	Standard MLP; batch normalization challenges in encrypted domain

Quadratic complexity manageable for **typical gait lengths (20-100 frames)**; **linear approximations** (Performer, Linformer) for **very long sequences** (ECVA) .

5.2 Occlusion-Aware Mechanisms

5.2.1 Attention-Based Feature Weighting

5.2.1.1 Learnable Occlusion Masks for Dynamic Feature Suppression **End-to-end occlusion estimation**:

Stage	Output	Supervision
Occlusion detection network	Per-pixel or per-part visibility score	Explicit mask annotation (if available) or implicit from recognition task
Feature masking	Element-wise multiplication of features with visibility	Gradient from recognition loss
Mask refinement	Progressive refinement through network depth	Learned update rules

Soft masking (continuous weights) **preserves gradient flow** vs. **hard masking** (binary threshold) (ECVA) .

5.2.1.2 Part Visibility Scoring Network **Explicit reliability estimation**:

```
Input: Part-specific feature vector + global context
↓
Visibility scorer: FC layers → sigmoid
↓
Output: Scalar visibility score per part [0,1]
```

Temporal consistency: **Visibility scores should vary smoothly across frames — enforced through temporal smoothing loss or recurrent estimation** (ECVA) .

5.2.1.3 Multi-Head Self-Attention with Occlusion-Aware Masking **Modified attention computation**:

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}} \odot M\right) V$$

where M is **occlusion mask** (0 for occluded positions, 1 for visible) applied **before softmax**.

Multiple heads enable diverse attention patterns: local (adjacent frames), cycle-level (same phase in previous cycles), part-based (corresponding parts across frames) (ECVA) .

5.2.2 Multi-Scale Feature Fusion

5.2.2.1 Pyramid Pooling for Scale-Invariant Representation

Level	Spatial Resolution	Information Captured	Occlusion Robustness
1×1	Global	Overall body proportions	High (coarse, robust to detail loss)
2×2	Quartered	Regional body part relationships	Moderate
4×4	Sixteenths	Fine-grained limb detail	Low (sensitive to occlusion)

Multi-level aggregation provides **redundancy**: **fine-scale loss compensated by coarse-scale preservation** (ECVA) .

5.2.2.2 Cross-Scale Feature Aggregation **Information flow between scales**:

Direction	Mechanism	Purpose
Bottom-up	Strided convolutions, pooling	Context aggregation
Top-down	Upsampling, lateral connections	Detail preservation with context guidance
Skip connections	Same-scale feature combination	Direct gradient flow, fine detail preservation

Feature Pyramid Network (FPN) architecture **naturally supports occlusion handling**: **coarse features guide completion of fine-scale occluded regions** (ECVA) .

5.2.2.3 Late Fusion of Appearance and Motion Cues

Stream	Input	Strengths	Occlusion Sensitivity
Appearance (GEI/shape)	Single-frame or aggregated silhouette	Rich shape detail; stable	High to shape-altering occlusion
Motion (optical flow/frame differences)	Temporal derivatives	Dynamic gait characteristics; occlusion detection cue	High to motion-breaking occlusion

Fusion strategy: **learned weighting based on estimated stream reliability**, or **concatenation with subsequent joint processing** (ECVA) .

5.3 Training Strategies

5.3.1 Loss Function Design

5.3.1.1 Triplet Loss with Hard Negative Mining

$$\mathcal{L}_{\text{triplet}} = \sum_{i}^{N} \left[\|f(x_i^a) - f(x_i^p)\|_2^2 - \|f(x_i^a) - f(x_i^n)\|_2^2 + \alpha \right]_+$$

Component	Selection Strategy	Effect
Anchor (x^a)	Random sample from identity	Reference point
Positive (x^p)	Different sample, same identity	Pull together
Negative (x^n)	Hardest negative in batch (semi-hard or hard)	Push apart most challenging confusers

Hard negative mining focuses training on **informative, difficult cases** — **critical for learning fine-grained discrimination** (ECVA) .

5.3.1.2 Center Loss for Intra-Class Compactness

$$\mathcal{L}_{\text{center}} = \frac{1}{2} \sum_{i=1}^{m} \|f(x_i) - c_{y_i}\|_2^2$$

Class centers c_{y_i} updated as **moving average of batch features**:

$$c_j^{(t+1)} = c_j^{(t)} - \beta \cdot \Delta c_j^{(t)}$$

Combined with triplet loss: inter-class separation + intra-class compactness (ECVA) .

5.3.1.3 Occlusion-Aware Consistency Regularization **Novel regularization for this project**:

$$\mathcal{L}_{\text{consistency}} = \sum_{i} \|f(x_i^{\text{clean}}) - f(x_i^{\text{occluded}})\|_2^2$$

Enforces embedding similarity across **different occlusion patterns of same subject** — **explicit training for occlusion invariance**.

Implementation: **synthetic occlusion generation** creating **paired samples** from **same base sequence** (ECVA) .

5.3.2 Data Augmentation

5.3.2.1 Random Synthetic Occlusion Generation

Parameter	Range	Effect
Occlusion probability	0.3-0.7 per sequence	Exposure to diverse occlusion levels
Position	Random or part-biased	Spatial variation
Size	10%-50% of silhouette area	Severity variation
Shape	Rectangle, ellipse, free-form	Realism (free-form from real occluders)

Dynamic application: **different random masks each epoch** prevents **overfitting to specific patterns** (ECVA) .

5.3.2.2 View-Angle Simulation Through Affine Transformation

Transformation	Parameter Range	Purpose
Rotation	$\pm 15°$	Viewpoint variation
Scaling	0.9-1.1	Distance simulation
Shear	± 0.1	Perspective effects
Translation	$\pm 5\%$ image size	Position jitter

Combined with occlusion: **joint robustness to view + occlusion variation** (ECVA) .

5.3.2.3 Temporal Resampling for Speed Variation

Technique	Implementation	Effect
Frame dropping	Random skip 1-2 frames	Faster apparent speed
Frame duplication	Random repeat 1-2 frames	Slower apparent speed
Linear interpolation	Subsample/interpolate to target length	Continuous speed change

Speed invariance critical for **cross-subject generalization** — **natural walking speed varies 2×** **across population** (ECVA) .

6. Implementation Framework

6.1 Development Environment

6.1.1 Core Libraries

Library	Version	Purpose	Key Features
PyTorch 2.0+	2.0.0	Primary deep learning framework	`torch.compile` for optimization; dynamic graphs; strong ecosystem
TensorFlow 2.12+	2.12.0	Alternative framework	Production deployment tools; Keras API
OpenCV	4.7.0	Image processing	Efficient preprocessing; GPU acceleration; video I/O
NumPy/SciPy	1.24.0/ 1.10.0	Numerical computation	Array operations; optimization; signal processing
Albumentations	1.3.0	Augmentation pipelines	Efficient transforms; bounding box/mask handling

Recommendation: **PyTorch 2.0+** for **research flexibility**; **TensorFlow** if **production deployment** (TensorFlow Serving, Edge TPU) is **priority** (ECVA) .

6.1.2 Specialized Tools

Tool	Purpose	Integration
MMDetection/ Detectron2	Pose estimation	Skeleton-guided occlusion recovery; part detection
Weights & Biases	Experiment tracking	Hyperparameter logging; result visualization; collaboration
TensorBoard	Alternative tracking	Local visualization; lightweight
ONNX Runtime	Deployment optimization	Cross-platform inference; hardware acceleration

6.2 System Integration

6.2.1 Modular Pipeline Architecture

Module	Components	Interfaces
Data Ingestion	Dataset loaders; augmentation; batching	Batched tensors with occlusion masks; metadata
Model Training	Network forward/backward; optimization; checkpointing	Model weights; training metrics; validation performance
Inference & Evaluation	Model application; metric computation; result aggregation	Rank-k accuracy; CMC curves; mAP; timing benchmarks
Visualization & Reporting	Attention maps; failure analysis; figure generation	Publication-ready plots; qualitative examples

6.2.2 Reproducibility Framework

Component	Implementation	Purpose
Configuration management	YAML/JSON files	Precise experiment specification; version control
Version control	Git + DVC	Code evolution tracking; large file (data, models) versioning
Containerization	Docker	Environment consistency; deployment portability

7. Experimental Analysis Plan

7.1 Evaluation Protocol

7.1.1 Standard Metrics

Metric	Definition	Target Performance
Rank-1 accuracy	% probes with correct identity at top-1	>85% on OccGait CO (vs. ~70% GaitPart baseline)
Rank-5 accuracy	% probes with correct identity in top-5	>95% on OccGait CO
CMC curve	Cumulative match characteristic	AUC >0.90
mAP	Mean average precision (multi-gallery)	>80%

7.1.2 Occlusion-Specific Analysis

Analysis Type	Method	Insight
Degradation curves	Accuracy vs. occlusion severity (size, duration)	Operational thresholds for deployment
Part contribution	Ablate individual parts; measure accuracy drop	Critical body regions for protection
Cross-type generalization	Train on CO, test on SO; vice versa	Transferability of occlusion handling

7.2 Comparative Benchmarking

7.2.1 Baseline Methods

Method	Characteristics	Expected Performance
GEI + SVM	Classical; no deep learning; no occlusion handling	~60% Rank-1 on OccGait CO
CNN-LSTM	Deep; temporal; occlusion-agnostic	~70% Rank-1 on OccGait CO

Table 47 – continued

Method	Characteristics	Expected Performance
GaitSet/ GaitPart	State-of-the-art; set-based/ part-based	~75% Rank-1 on OccGait CO (ECVA)
GaitMoE	SOTA with occlusion focus; mixture-of-experts	~80% Rank-1 on OccGait CO (ECVA)

7.2.2 Ablation Studies

Component	Ablation	Expected Effect
Attention mechanism	Remove; replace with uniform weighting	Validate attention contribution
Temporal fusion	Single-frame processing only	Quantify temporal information value
Occlusion detection	Use ground-truth masks vs. learned	Estimate detection error impact
Part-based processing	Holistic features only	Validate part decomposition benefit

7.3 Statistical Validation

Aspect	Method	Confidence
Cross-validation	Subject-independent splits; view-stratified	Unbiased generalization estimate
Significance testing	Paired t-test across multiple runs	$p < 0.05$ for claimed improvements
Uncertainty quantification	Bootstrap confidence intervals	95% CI for reported metrics

8. Research Paper Structure

8.1 Core Contributions

Contribution	Description	Evidence
Architectural innovation	Attention-fusion integration for occlusion-aware gait recognition	Ablation studies; comparison with GaitMoE
Evaluation framework	Systematic analysis across occlusion types, severity, cross-dataset	Comprehensive tables and figures
Open-source implementation	Public code with reproducible benchmarks	GitHub repository; documentation

8.2 Anticipated Results

Category	Result	Significance
Quantitative	>10% Rank-1 improvement over GaitPart on OccGait CO	Demonstrates occlusion handling effectiveness
Qualitative	Attention visualizations showing robust focus on visible parts	Interpretability; failure mode understanding
Practical	Inference timing <100ms per sequence on standard GPU	Deployment feasibility

8.3 Publication Targets

Venue Tier	Specific Venues	Positioning Strategy
Top-tier conferences	CVPR, ICCV, ECCV workshops	Emphasize practical impact; real-world problem solving
Specialized journals	IEEE T-BIOM, Pattern Recognition	Full methodological detail; extensive evaluation
Application venues	IEEE T-CSVT, Neurocomputing	Surveillance deployment focus; efficiency emphasis